resolution to the quantum measurement puzzle, may or may not continue to treat each direction in time equally. Let's consider the implications of each possibility.

If time symmetry persists (as I suspect it will), all of the reasoning and all of the conclusions of the last chapter can be carried over with little change to the quantum realm. The core physics that entered our discussion of time's arrow was the time-reversal symmetry of classical physics. While the basic language and framework of quantum physics differ from those of classical physics—wavefunctions instead of positions and velocities; Schrödinger's equation instead of Newton's laws—time-reversal symmetry of all quantum equations would ensure that the treatment of time's arrow would be unchanged. Entropy in the quantum world can be defined much as in classical physics so long as we describe particles in terms of their wavefunctions. And the conclusion that entropy should always be on the rise—increasing both toward what we call the future and toward what we call the past—would still hold.

We would thus come to the same puzzle encountered in Chapter 6. If we take our observations of the world right now as given, as undeniably real, and if entropy should increase both toward the future and toward the past, how do we explain how the world got to be the way it is and how it will subsequently unfold? And the same two possibilities would present themselves: either all that we see popped into existence by a statistical fluke that you would expect to happen every so often in an eternal universe that spends the vast majority of its time being totally disordered, or, for some reason, entropy was astoundingly low just following the big bang and for the last 14 billion years things have been slowly unwinding and will continue to do so toward the future. As in Chapter 6, to avoid the quagmire of not trusting memories, records, and the laws of physics, we focus on the second option—a low-entropy bang—and seek an explanation for how and why things began in such a special state.

If, on the other hand, time symmetry is lost—if the resolution of the measurement problem that is one day accepted reveals a fundamental asymmetric treatment of future versus past within quantum mechanics—it could very well provide the most straightforward explanation of time's arrow. It might show, for instance, that eggs splatter but don't unsplatter